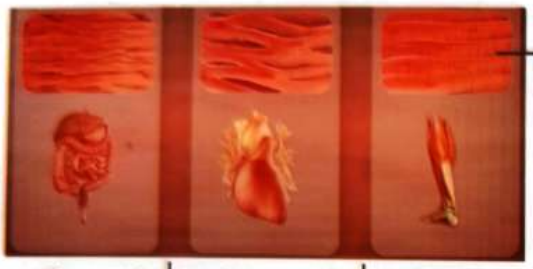
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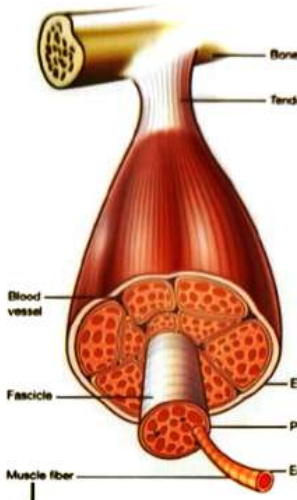
SKELETAL MUSCLES

----- Active space --



Smooth muscle | Cardiac muscle | Skeletal muscle

→ Function : Contraction & Relaxation



- Epimysium → Connective tissue
- Perimysium → Overall outer covering
- Endomysium → Covering of fascicle
- Muscle fiber → Covering of muscle fiber

Has cell membrane : Sarcolemma

Sarcomere

00:03:00

Functional unit of skeletal muscle : Sarcomere.

Proteins :

- Actin & myosin.

Actin filament/thin filament.



→ G-actin

Myosin head region : ATPase activity

Myosin → 2 heavy chains, 4 light chains

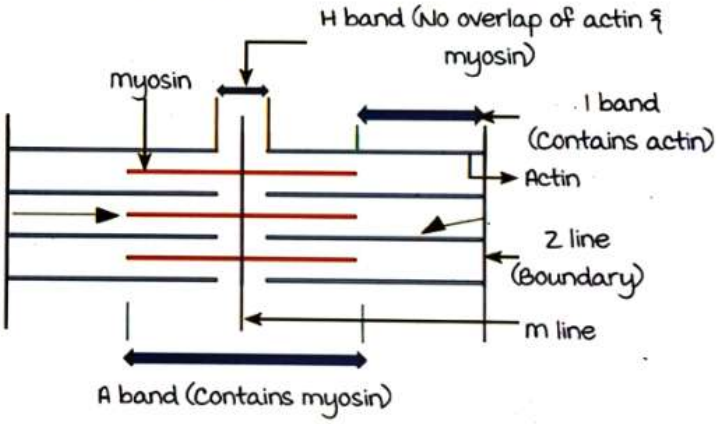
- Thick filament.
- 1.6 µm fixed length.

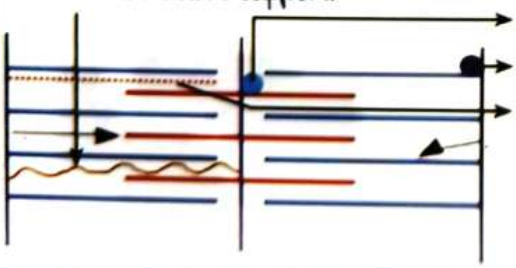
- α-actinin, myomesin, Nebulin & Titin.

Titin :

- Extends from Z line to M line.
- Spring like appearance.
- Provides elastic support.

Microscopic appearance :





- Myomesin (Attaches myosin to M line)
- α-Actinin (Attaches actin to Z line)
- Nebulin :
  - Follows entire length of actin.
  - Forms a molecular ruler.
  - Regulates length of actin.

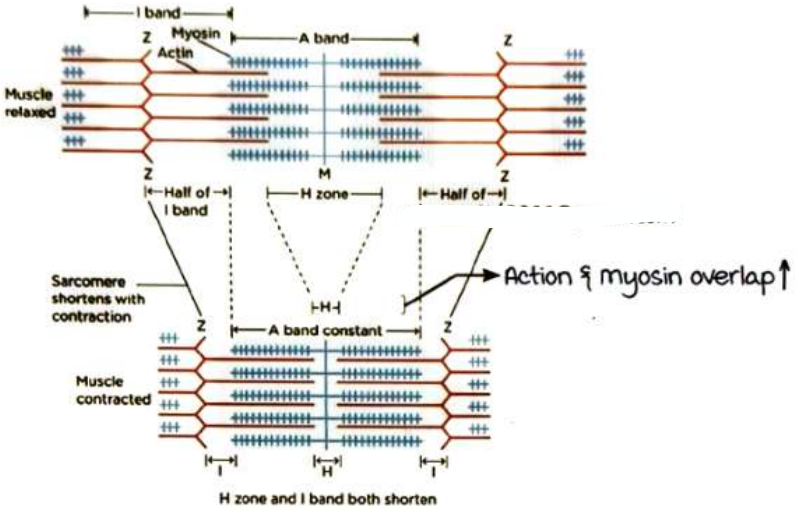


1 sarcomere = Area between 2 Z - lines

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----- Active space -----

Mechanism of contraction :



Skeletal muscle proteins

00:16:40

Types

Contractile proteins
- Actin.
- Myosin.

Supporting proteins :
- Desmin.
- Titan.
- Dystrophin.

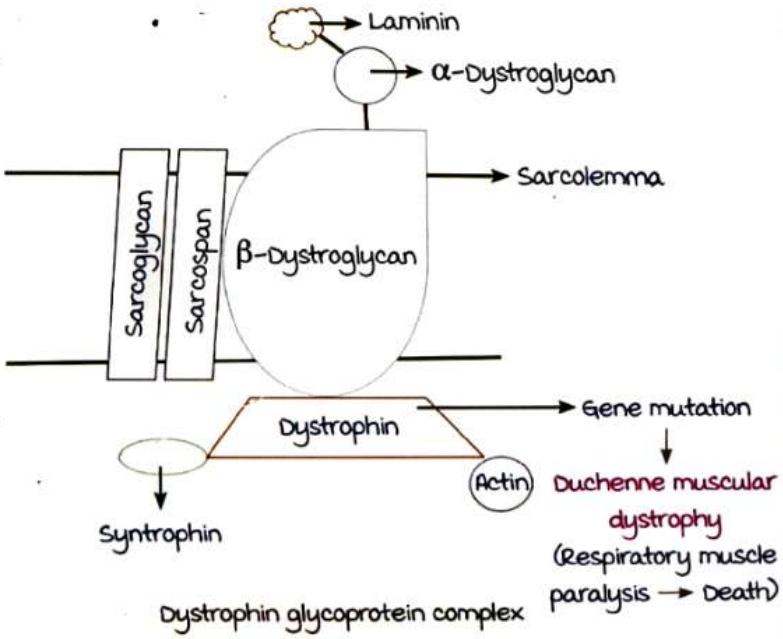
Regulatory proteins :
- Troponin C.
- Troponin I.
- Troponin T.

Relaxation Proteins :
- Sarcoplasmic reticulum Ca^{2+} ATPase (SERCA).
  - Ca^{2+} channel protein.
  - Activation of SERCA → muscle relaxation.

SUPPORTING PROTEINS :

Desmin :
- Type of Intermediate filament.
- Tumor marker for Sarcomas.
- Connects Z line to sarcolemma.

Titin :
- Giant/largest protein.
- Provides elasticity & passive tension.
- Scaffolding protein.



Dystrophin :
- Present as dystrophin glycoprotein complex.
- Connects sarcolemma to actin.

Feedback

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REGULATORY PROTEINS :

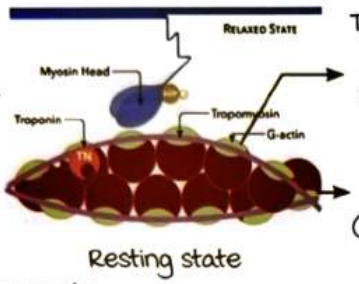
----- Active space -----

1. Troponin :

Subunits

- Troponin C (T_C).
  - Binds to Ca^{2+}.
- Troponin I (T_I).
  - Inhibitory.
- Troponin T (T_T).
  - Binds to tropomyosin.

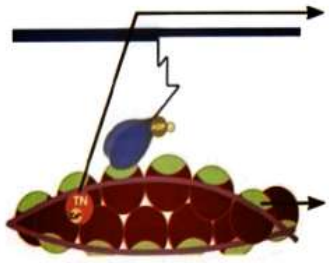
a. Tropomyosin :



Resting state

Tropomyosin covers myosin binding sites on actin

Tropomyosin (Rope-like protein)



Contraction state

Ca^{2+} binds to troponin → myosin binding site is exposed → Actin can take part in muscle contraction.

Synaptic Events in Muscle Contraction

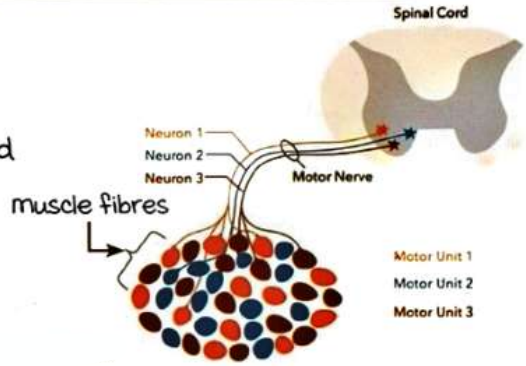
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Excitation- contraction coupling :

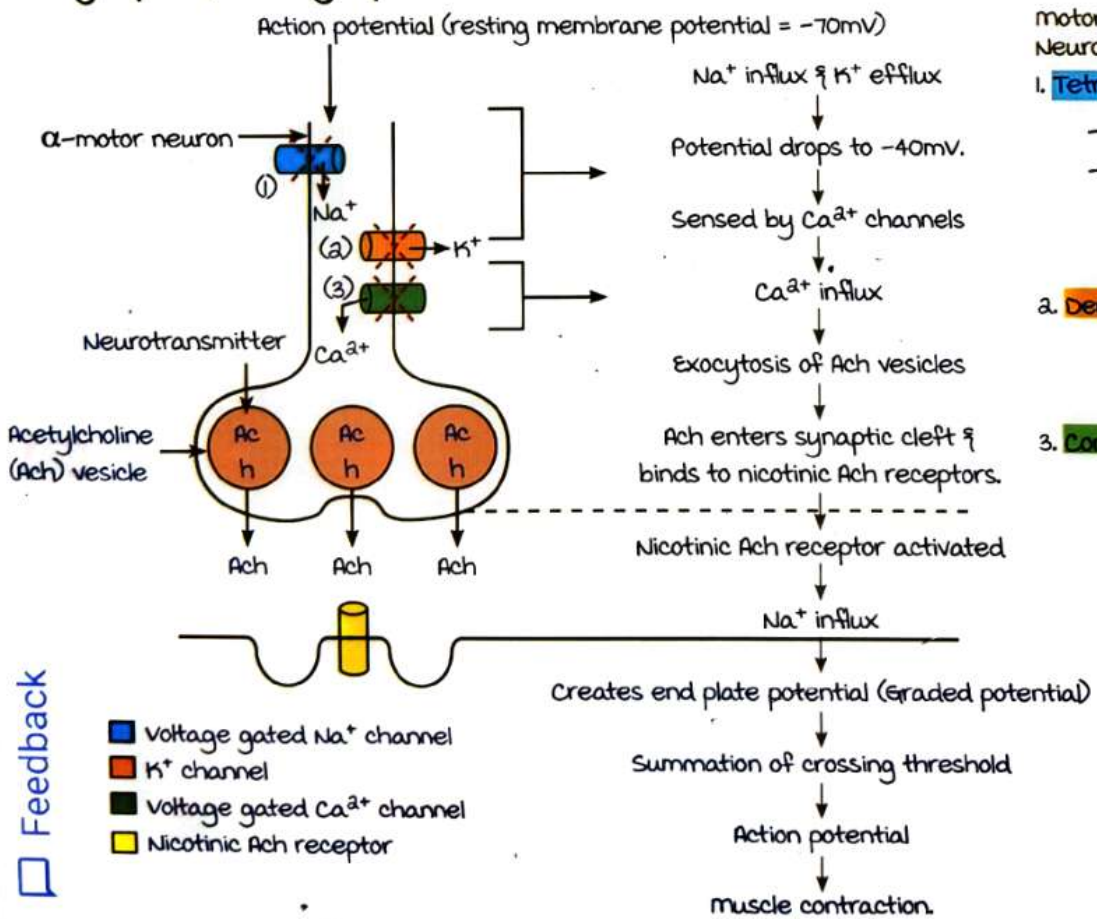
Skeletal Muscle

↓ Action potential : Electrical event } Ca^{2+} mediated coupling

Contraction : Mechanical event



Pre Synaptic & Post Synaptic events :



Na^+ influx & K^+ efflux

↓ Potential drops to -40mV.

↓ Sensed by Ca^{2+} channels

↓ Ca^{2+} influx

↓ Exocytosis of Ach vesicles

↓ Ach enters synaptic cleft & binds to nicotinic Ach receptors.

↓ Nicotinic Ach receptor activated

↓ Na^+ influx

↓ Creates end plate potential (Graded potential)

↓ Summation of crossing threshold

↓ Action potential

↓ Muscle contraction.

motor units

Neuromuscular toxins/blockers :

1. Tetradotoxin :
   - Present in pufferfish
   - Leads to muscle weakness/paralysis → Death.
2. Dendrotoxin :
   - Present in black mamba snake.
3. Conotoxin :
   - Present in snails.

Feedback

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----- Active space -----

Disorders of neuromuscular transmission :

Myasthenia Gravis	Lambert Eaton Myasthenic Syndrome
Autoantibodies against Nicotinic Ach receptors.	Autoantibodies against Voltage gated Ca^{2+} channels.
Postsynaptic	Presynaptic

Note :

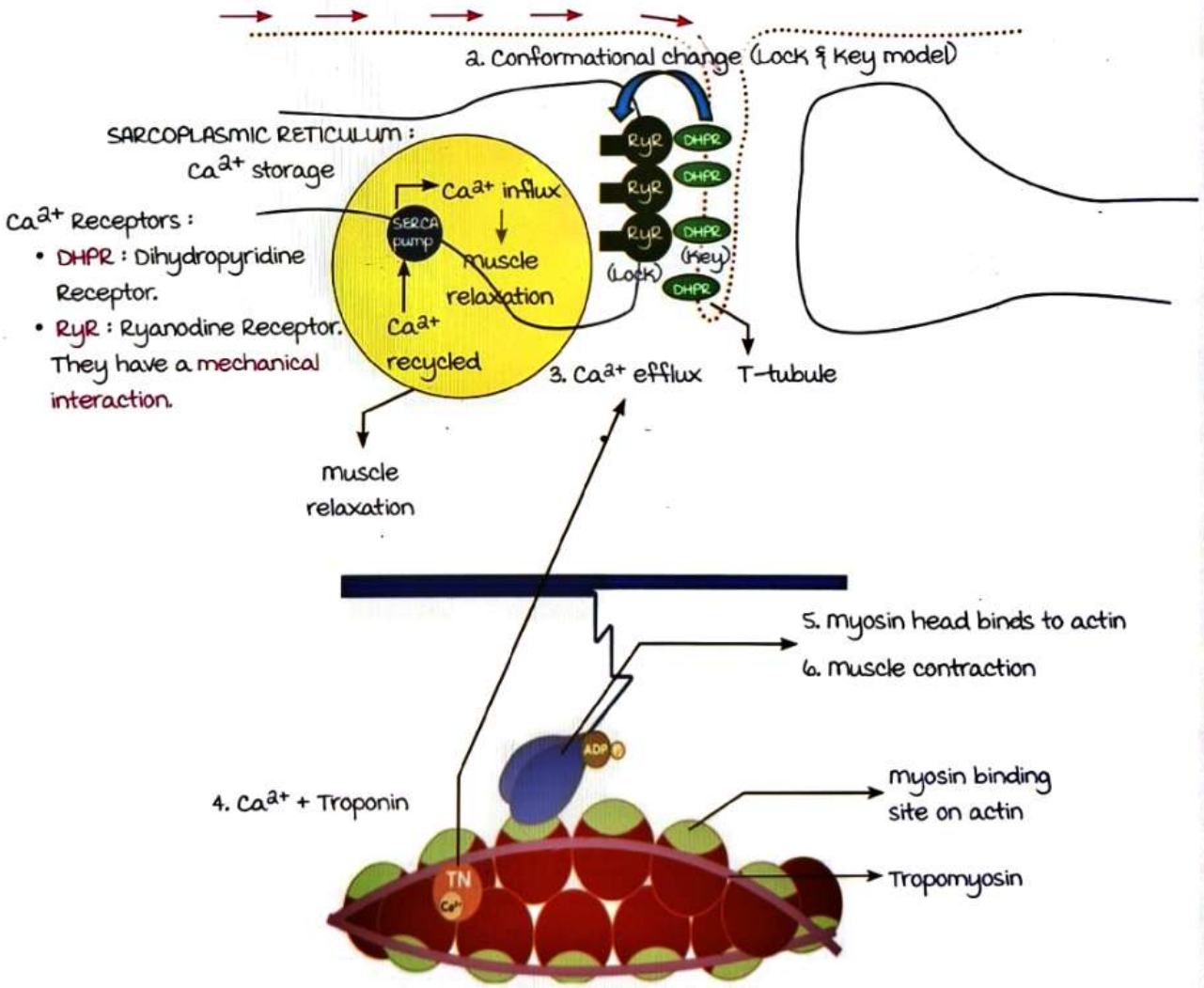
- Miniature end plate potential/quantal Due to spontaneous release of Ach :

Muscle contraction

00:44:58

SEQUENCE OF EVENTS :

1. Action potential.



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Applied Aspect :

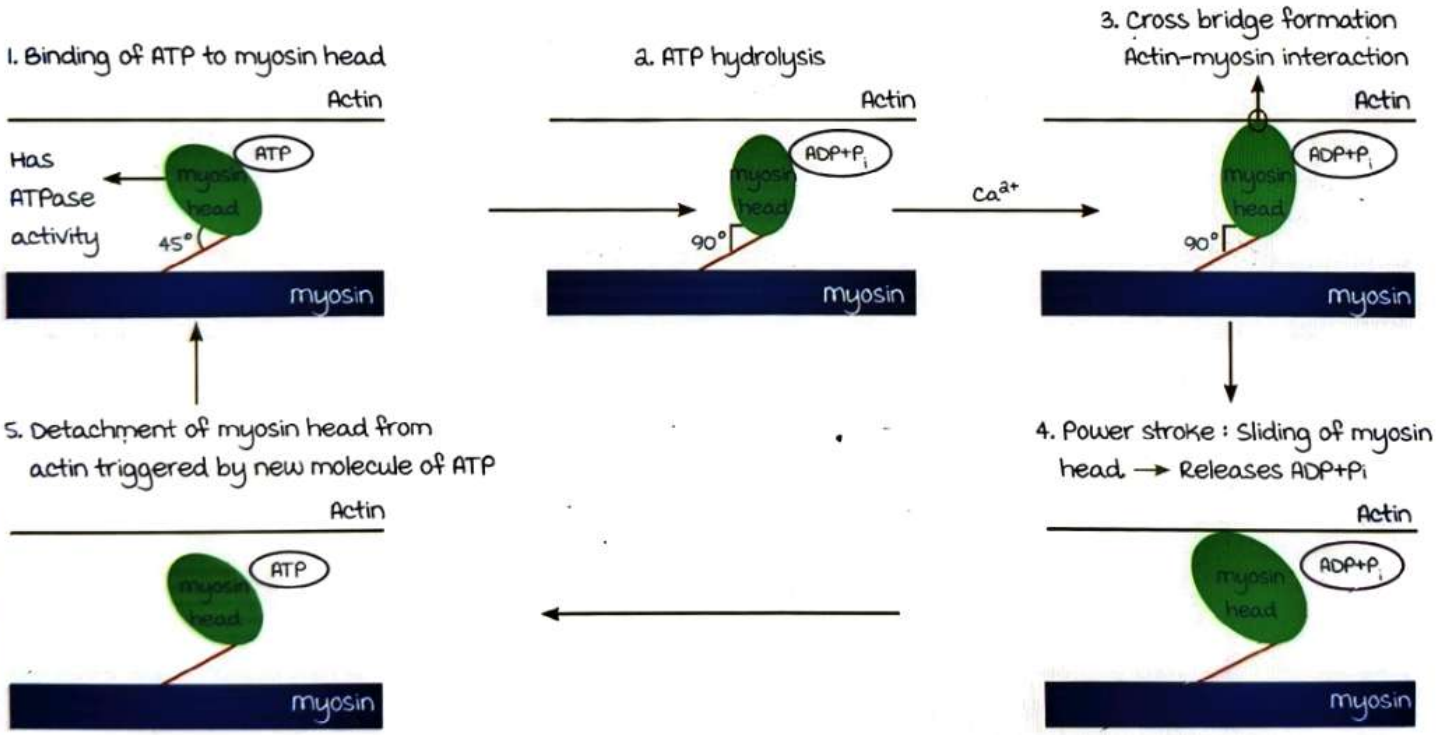
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1. DHPR blockers :
   - Calcium channel blockers → Eg : Nifedipine.

2. RyR blockers :
   - Eg : Dantrolene Sodium.
   - used in malignant hyperthermia.
     - MOA of malignant hyperthermia: Overactive RyR (↑↑Ca^{a+} release) → Sustained muscle contraction → ↑↑ body heat.

3. Brody's myopathy :
   - SERCA pump : Coded by ATP 2AI gene.
     - ATP 2AI Mutation → Affects muscle relaxation → muscle cramps following exercise.

MOLECULAR MECHANISM :



Cross bridge cycling + Power stroke → Enhanced muscle contraction.

Feedback